LLM Judges Don't Need Psychometrics or Correlation: Just Match the Human Score Distribution

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Abstract

LLM judges are increasingly the measuring instrument for AI scientists, yet a judge is usually validated by the correlation (Pearson, Spearman, or Kendall) between its scores and human scores, sometimes supplemented with validity frameworks borrowed from psychometrics. We argue that neither is needed. A judge is a conditional distribution over scores, $p_\theta(r \mid y, y^*)$, and the question "does this judge agree with humans?" is the standard machine-learning question of whether one distribution matches another. Correlation summarizes only one property of that relationship: linearity for Pearson, ordering for rank correlations. A judge can therefore score well while its score distribution differs sharply from the human one. We propose fitting and selecting judges by minimizing $D_{\mathrm{KL}}(p_{\mathrm{human}} \,\|\, p_\theta)$, reusing the mature optimization toolbox built for this objective, and reporting normalized squared error for interpretability. We give a synthetic counterexample and outline a planned case study on judging theorem-proving agents.

Introduction

Automated research systems are increasingly scored by LLM judges, so the reliability of the judge bounds the reliability of every reported gap between systems. The dominant recipe for validating a judge is simple. Collect human ratings, collect judge ratings on the same items, and report their correlation (Zheng et al. 2023; Liu et al. 2023). When that feels insufficient, practitioners often turn to psychometrics for reliability and validity criteria.

This paper takes the position that both moves are unnecessary and that correlation is actively misleading as an objective. If humans have a "taste" (a distribution over scores they assign to an output), then the judge should match that taste distribution. Matching distributions is the core operation of modern machine learning. Maximum likelihood minimizes a KL divergence (Kullback and Leibler 1951), and it is the objective large language models are trained on. We do not need a new instrument theory. We need to use the one we already have.

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Contributions. (i) We formalize a judge as a conditional score distribution and contrast the correlation objective with a KL objective (Proposal section). (ii) We show a synthetic case where two judges have near-identical correlation with humans but very different KL (Synthetic Counterexample section). (iii) We outline a planned evaluation on judges of theorem-proving agents (Case Study section).

What Correlation Measures

Pearson's r (Pearson 1895) measures only the strength of a *linear* relationship. Spearman's ρ (Spearman 1904) and Kendall's τ (Kendall 1938) measure only agreement in *ordering*. Each statistic is a single summary of a joint distribution, and it is well known that very different joint distributions can share the same summary (Anscombe 1973).

For judges, the consequence is concrete. Any strictly increasing transform of a judge's scores leaves rank correlation unchanged, so a judge that compresses every score into $\{4, 5\}$ while preserving order looks as good as one that reproduces the human scale. Correlation also discards how much humans disagree on each item, which is exactly the information that describes their taste. Optimizing a judge for correlation therefore optimizes for one moment-like property while leaving the rest of the distribution unconstrained, and that invites spurious agreement.

Psychometric frameworks address real concerns, but they were designed for settings where the instrument cannot be fit directly to data. An LLM judge can be. Once the target is a distribution, its fit can be measured and optimized directly.

Proposal: Judges as Distributions

Let y be an agent's output, y^* a gold reference, and r a score, for example $r \in \{0, \ldots, 5\}$. A judge is a distribution $p_\theta(r \mid y, y^*)$, where θ is anything we can search over: prompt, rubric, sampling temperature, calibration parameters, or weights. Current practice amounts to

$$\theta^\star = \arg\max_\theta \ \mathrm{corr}\left(r_\theta, r_{\mathrm{human}}\right). \tag{1}$$

We instead advocate

$$\theta^\star = \arg\min_\theta \mathbb{E}_x\left[D_{\mathrm{KL}}\left(p_{\mathrm{human}}(\cdot \mid x) \,\|\, p_\theta(\cdot \mid x)\right)\right]. \tag{2}$$

where $x = (y, y^*)$. With several raters per item, p_{human} is the per-item empirical histogram, and Eq. 2 is equivalent up

to a constant to the cross-entropy of human ratings under the judge. With one rating per item, a pooled variant compares marginal histograms. That variant is weaker but cheap. Additive smoothing keeps the divergence finite. Because Eq. 2 is the standard ML objective, it can be minimized with standard tools such as Adam (Kingma and Ba 2015), or by black-box search over prompts.

Reporting. KL is a good objective but not an intuitive number to report. For bounded scores we propose also reporting mean squared error on scores normalized to $[0, 1]$. Unlike exact-match accuracy, it penalizes a judge that says 0 when humans say 5 far more than one that says 4.

A Synthetic Counterexample

We draw $n = 2000$ human scores uniformly from $\{0, \ldots, 5\}$. Judge A adds noise in $\{-1, 0, 1\}$ and clips the result. Judge B outputs 5 if the human score is at least 3 and 4 otherwise, so it preserves order but squashes the scale. KL is computed between Laplace-smoothed marginal histograms (seed 0; script in the supplementary repository).

Judge	Pearson	Spearman	KL	nMSE
A (noisy faithful)	0.900	0.900	0.001	0.023
B (squashed)	0.877	0.878	3.420	0.216

Table 1: Correlation barely separates the two judges. KL and normalized MSE do.

Planned Case Study: Judging Theorem-Proving Agents

We are collecting human ratings of whether theorem-proving agents produce good formal theorems relative to a reference (y^*). We plan to (i) fit LLM judges by Eq. 2 and by correlation maximization, (ii) compare them on held-out items using KL, normalized MSE, and correlation, and (iii) study how many human-rated pairs are needed for the KL estimate to stabilize. No results are reported here.

Counterarguments and Limitations

"Correlation-validated judges have worked so far." Progress despite a flawed metric does not make the metric right. At minimum, matching the distribution stops us from optimizing a quantity we do not care about. *"With enough data any metric works."* Not necessarily. A metric that ignores most of the distribution can be satisfied by degenerate judges no matter how much data is available. *"KL needs many ratings per item."* This is true for per-item KL. The pooled variant and smoothing are partial remedies, and sample complexity is part of our planned study. Finally, it is possible that in practice the choice of objective changes little. That is an empirical question, and the case study is designed to answer it.

Conclusion

A judge is a distribution, so validate it the way we validate distributions. If humans have a taste, match the taste distribution.

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